Weight-Loss Attempts, Behaviors, and Weight Change Among U.S. Adults

Which everyday behaviors predict one-year weight change, beyond demographics and starting BMI?

Thomas Duong Maggie Ha William Coleman

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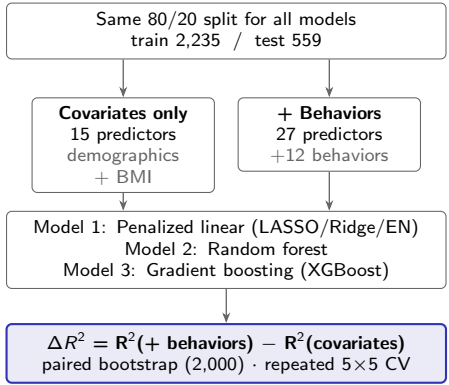
Data: NHANES, August 2021 – August 2023 ($n = 2{,}794$ adults)

Research Question

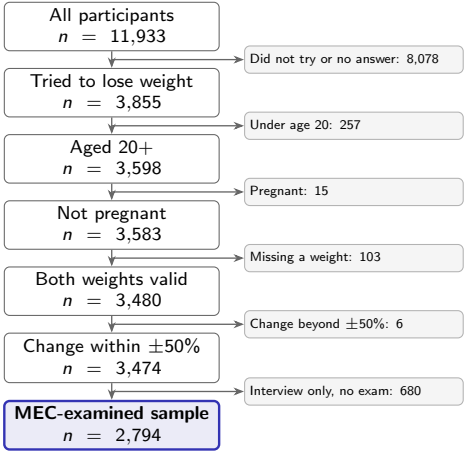
Which behaviors best predict **one-year % weight change** among U.S. adults who tried to lose weight, **beyond demographics and starting BMI**?

- Data: **NHANES 2021–2023**, 9 public files merged, $n = 2{,}794$ adults.
- Outcome: $\%\Delta W = (W_{\text{now}} - W_{1\text{ yr ago}})/W_{1\text{ yr ago}} \times 100$ (negative = loss)

Design: every model is fit twice



Data cleaning: 11,933 NHANES participants → 2,794 in the analysis



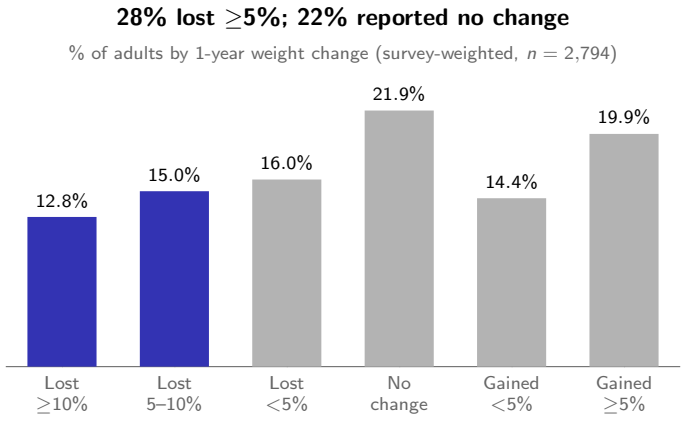
- **Merged 9 files** on `SEQN`: demographics, weight history, activity, diet, sleep, smoking, alcohol, PHQ-9, body measures.
- **Dropped:** child-only, admin and redundant columns, and measured body size (outcome leakage).
- **Removed columns with more than 30% missing data** (3 columns: time in U.S., community meals delivered, meals at a senior center).

Features: 15 covariates + 12 behaviors

Missing: ≈12% for PHQ-9, drinks/week, income; <1.5% elsewhere

Model 1: median imputation + missingness flags Models 2–3: native NaN handling in trees

Data Exploration



Model 1 – Penalized linear regression: behaviors add ≈0.006 R^2

Method

- **LASSO** (L1, selects), **Ridge** (L2, shrinks), **Elastic Net** (L1+L2).
- Pipeline: train-median imputation + missingness flags → standardize → fit.
- Penalty chosen by 10-fold CV on training set.
- Stability: 200 bootstrap resamples, % of times each predictor is kept by LASSO.

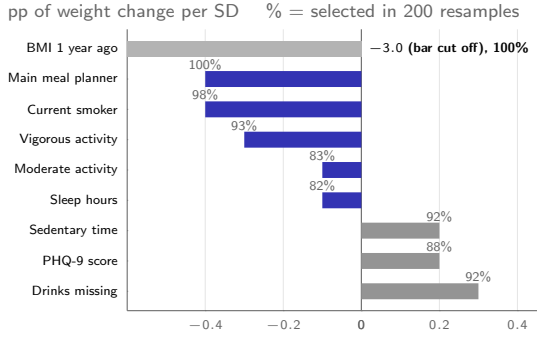
Test-set results

	Covariates R^2	+ Behaviors R^2	RMSE
LASSO	0.074	**0.080**	8.71
Ridge	0.072	0.079	8.72
Elastic Net	0.075	**0.080**	8.71

Mean-only baseline: $R^2 = -0.002$, RMSE 9.09

ΔR^2 test: +0.005 to +0.008, **95% CI includes 0**. Repeated CV: +0.008; behaviors better in 20–21 of 25 folds.

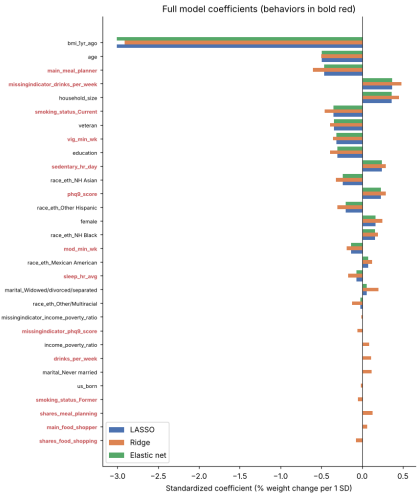
LASSO standardized coefficients (bootstrap median)



Model 1 – Figures: penalties agree; predictions stay near the mean

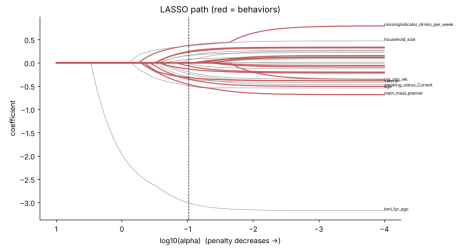
Standardized coefficients, all 27 predictors

LASSO, Ridge and Elastic Net bars nearly overlap; behaviors in red



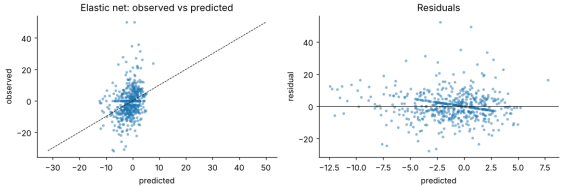
LASSO path

BMI enters first and stays largest; behaviors (red) enter near the chosen penalty (dashed line)



Elastic Net on the test set: observed vs. predicted, residuals

Predictions span about −12 to +8%; observed change spans ±40%



Model 2 – Random forest: moderate activity is the one robust behavior

Method

- 1,000 trees; native missing-value handling (no imputation).
- Randomized search (30 × 5-fold CV) over depth, leaf size, features, sample fraction.
  Chosen (+ behaviors): depth 6, min leaf 19, 70% features.
- Interpretation: test-set **permutation importance**, **SHAP**, and a **Boruta** shadow-feature test (100 rounds, Bonferroni).

Test-set results

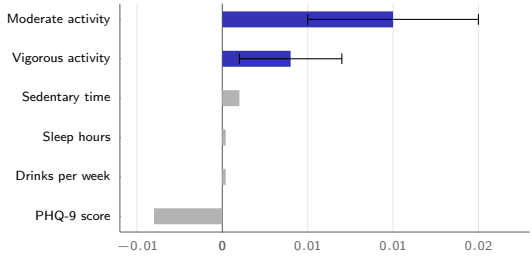
	Test R^2	RMSE	CV R^2
Covariates only	0.071	8.75	0.122
+ Behaviors	**0.074**	**8.74**	**0.128**
4 predictors only†	0.079	–	0.120

†BMI, age, income, moderate activity

ΔR^2 test: +0.003 [−0.015, 0.021]; CV +0.006, better in 19/25 folds.
Boruta keeps only 6 of 27: BMI, age, income, PHQ-9, moderate activity, sleep.

Permutation importance of behaviors (test ΔR^2)

drop in test R^2 when shuffled; bars = 95% CI where reported

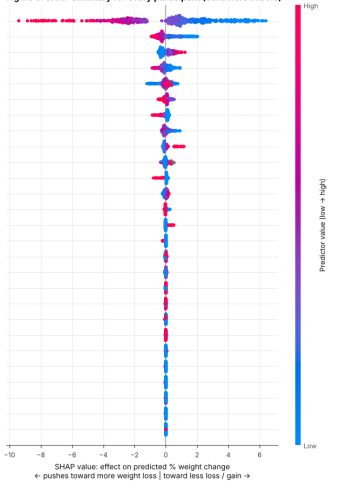


Takeaway: BMI alone is worth $\Delta R^2 \approx 0.20$ (12× next). PHQ-9 and sleep have large SHAP and pass Boruta but *don't* help on the test set → the forest partly fits noise with them.

Model 2 – Figures: what the forest uses, and how

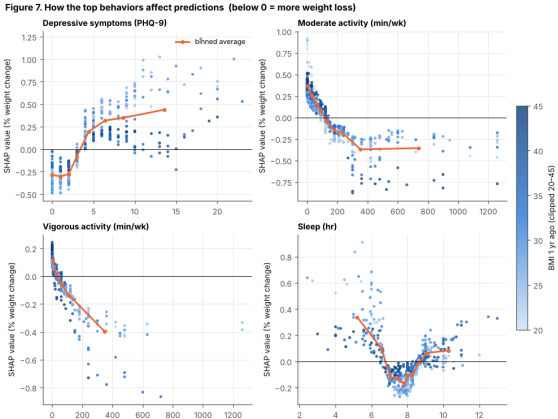
SHAP summary (test set)

High starting BMI (red) pushes predictions toward loss



SHAP dependence, top 4 behaviors

Below 0 = more loss. PHQ-9 rises after ≈3; activity benefit flattens after ≈300 min; sleep is U-shaped around 7–8 h



Model 3 – Gradient boosting (XGBoost): largest gain, non-linear shapes

Method

- Sequential shallow trees, each correcting earlier errors; native NaN handling.
- Randomized search (60 × 5-fold CV) biased to **strong regularization**: chosen: depth 3, 507 trees, lr 0.012, min child 39, λ=2.4, α=4.5, γ=1.1
- Interpretation: test-set permutation importance, grouped permutation, TreeSHAP.

Test-set results

	Test R^2	RMSE	CV R^2
Covariates only	0.065	8.78	0.127
+ Behaviors	**0.075**	**8.74**	**0.138**

ΔR^2 test: +0.010 [−0.015, 0.033]; CV +0.011, better in **22/25** folds.
Shuffle all 12 behaviors: −0.024 R^2 vs. all 15 covariates: −0.246.

Behavior importance: permutation vs. SHAP

Behavior	Perm.	|SHAP|	Shape (SHAP)
Moderate activity	**0.014***	0.41	▼ more loss; flat >150–300 min
PHQ-9	−0.004	0.33	▲ less loss if score ≥5
Sedentary time	0.007	0.28	▲ less loss above 5 h/day
Sleep hours	−0.001	0.25	▲ less loss under 6.5 h
Vigorous activity	0.006	0.24	▼ more loss
Current smoker	0.001	0.18	▼ more loss
Main meal planner	0.000	0.17	slightly ▼ more loss

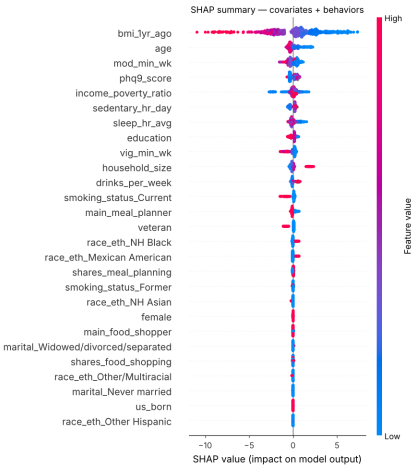
*95% CI [0.004, 0.025] excludes 0; all other behaviors' CIs include 0.

BMI 1 yr ago: permutation $\Delta R^2 = 0.22$, mean |SHAP| = 2.5 pp. Strongest behavior interactions are with BMI (moderate activity × BMI, PHQ-9 × BMI). Shapes are **descriptive**: boosting did not beat the linear models.

Model 3 – Figures: same ranking as the forest, with clearer thresholds

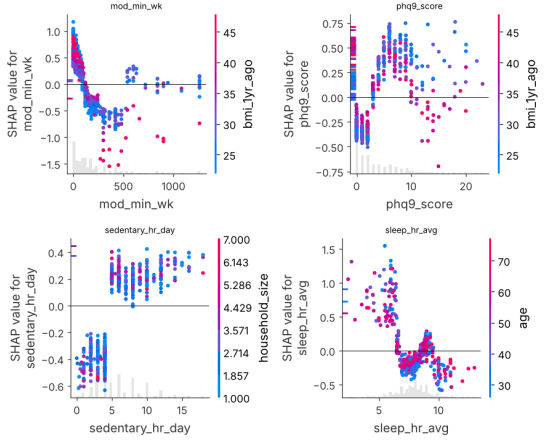
SHAP summary, covariates + behaviors

Moderate activity and PHQ-9 are the top behaviors after BMI and age



SHAP dependence: moderate activity, PHQ-9, sedentary time, sleep

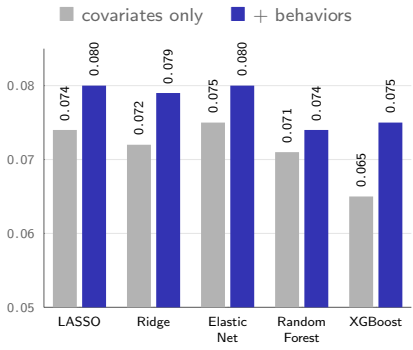
Activity: steep early benefit, then a plateau. Sedentary: a step near 5 h/day. Sleep: penalty under ≈6.5 h



Comparison

All models have $R^2 \approx 0.07$–0.08; behaviors always help a little.

Test R^2 by model and predictor set



Does adding behaviors help?

	Test ΔR^2 [95% CI]	CV	Folds
LASSO	+0.005 [−0.010, 0.021]		
Ridge	+0.008 [−0.011, 0.028]	+0.008	20–21
EN	+0.005 [−0.010, 0.021]		
RF	+0.003 [−0.015, 0.021]	+0.006	19
XGBoost	+0.010 [−0.015, 0.033]	+0.011	22

- **Consistent direction:** every model improves; behaviors win in 76–88% of CV folds.
- **Not significant:** every test-set CI includes 0.

Comparison: the three models agree on which behaviors matter

Behavior	Direction	M1 LASSO (coef, % kept)	M2 Random forest	M3 XGBoost
Moderate activity	▼ more loss	−0.1, 83%	**perm. 0.010***, Boruta ✓	**perm. 0.014***, flat past 150–300 min
Vigorous activity	▼ more loss	−0.3, 93%	perm. 0.004*	perm. 0.006
Main meal planner	▼ more loss	−0.4, 100%	–	small SHAP
Current smoker	▼ more loss	−0.4, 98%	–	small SHAP
Sedentary time	▲ less loss	+0.2, 92%	perm. 0.001	step above ≈5 h/day
PHQ-9 symptoms	▲ less loss	+0.2, 88%	Boruta ✓, high SHAP	less loss once ≥5
Short sleep	▲ less loss	−0.1/h, 82%	Boruta ✓	less loss under 6.5 h
BMI 1 year ago	▼ more loss	−3.0, 100%	perm. 0.20	perm. 0.22

* test-set 95% CI excludes 0. "–" = not among reported top behaviors. Coefficients in pp per SD; permutation = drop in test R^2.

Moderate activity is the only behavior with **test-set importance clearly above zero** in both tree models. Linear models favor meal planning and smoking; trees favor activity, mood and sleep, with **non-linear thresholds**.

Conclusion: BMI drives prediction; behaviors add a little

Findings

1. **Behaviors add a small, consistent gain** ($\Delta R^2 \approx 0.003$–0.011) across all models, but no test-set CI excludes zero.
2. **Starting BMI dominates**: $\approx 6\times$ any other predictor in linear models, $\Delta R^2 \approx 0.20$ in trees. Higher BMI $\rightarrow$ greater % loss.
3. **Most robust behavior: moderate activity**, with benefit leveling off around 150–300 min/week. Also: vigorous activity, meal planning (▼ more loss); sitting >5 h/day, PHQ-9 ≥ 5, sleep <6.5 h (▲ less loss).
4. **Simple beats complex**: linear $\approx$ trees ($R^2 \approx 0.08$); a 4-predictor forest matches the 27-predictor one.

Starting BMI matters more than self-reported habits, but staying **moderately active** is the habit most consistently linked to losing weight.